



Eating meat without killing the chicken

British scientists have joined the race to produce meat grown in the lab rather than reared in farms. <https://digit.in/apr19-19>



"Hole punch cloud" spotted

Locals were mystified after a strange 'Whirlpool Hole' in the Sky Spotted Over UAE and Oman. <https://digit.in/apr19-20>

la Lobe. The soil here is black in colour because of the rich iron content in the original magma outpourings. The soil also has a high capacity for retaining water, and is ideal for agriculture as the requirement for irrigation is minimal. Further east, there are subsurface Deccan Traps, all the way to the east coast. There are Deccan rocks on the surface or Rajahmundry, in Andhra Pradesh. Together, the rocks record the world's largest lava flow, from the Deccan Traps to Rajahmundry, a distance in excess of 1,500 kilometers. The lava made this journey in eight phases, flowing into the Bay of Bengal. At that time, the Rajahmundry area was beneath an ancient ocean, and marine fossils have been found in the basalt rocks. With increasing volcanic activity, the number and variety of marine fossils have been found to decrease. The ancient flow roughly tracks the route of the Godavari river today. The solidified flood basalt rocks form a layer that is 2,000 meters thick, and cover an area of 500,000 square kilometers. They are the among the largest volcanic features on the Earth. Originally, before they were eroded or buried under the surface, the Deccan traps covered a region of 1,500,000 square kilometers. Geologists have not been able to accurately ascertain how much of the lava flows are underwater, and how much have been eroded.

DECCAN VOLCANISM

About 66 million years ago, the continent of India was making its way to Asia, across the Indian ocean. India was roughly equidistant from Africa, Antarctica and Australia. This was the setting for one of the most massive volcanic eruptions in Earth's history. Now such volcanic eruptions are fed by convection currents of magma within the mantle of the Earth, a region so hot that the rock is in a liquid form, beneath the solid outer crust. Like thermals that rise in the atmosphere of the Earth, mantle plumes transport material from the very edges of Earth's core, all the way to the surface. Off the coast of Madagascar is a small island owned

by France, called Réunion. Beneath the island is the tail end of a mantle plume that feeds one of the most active shield volcanoes on Earth – the Piton de la Fournaise. The last eruption was in July 2017. Initially, geologists believed that the head end of this plume was responsible for the Deccan Volcanoes. New research suggests something even more incredible.

The size of the Deccan Traps today indicates the sheer amount of material that was ejected from the innards of the Earth. A single plume was not sufficient to explain the sheer amount of material that flowed out. Researchers from the University of Quebec in

accepted. The challenging hypothesis is that the volcanic activity was not because of mantle plumes at all, but because of plate tectonics, or the way in which the plates that make up the crust of the Earth move about.

The timing and periodicity of the eruptions are also a matter of research. The eruptions took place in phases, spread over millions of years. The process of finding out the periodicity and volume of the lava flows is to geologically fingerprint the layers of rocks, and study their size and order. The characteristics of these layers are uniform throughout the Deccan Traps. The activity is known as stratigraphy,

and divides the layers into formations, which indicate the time periods of the activity. The latest studies indicate that most of the volcanic activity happened within a period of one million years, the second and largest of three pulses.

Accompanying the volcanic eruptions was a massive outpouring of gasses, including carbon dioxide and sulphur dioxide. These are greenhouse gasses that drastically changed the climate of the entire world at that time. The particulate matter in the atmosphere initially shielded the surface from



The area of the Deccan Traps

Canada that ran computer simulations going back 70 million years, showed something interesting. At the time of the Deccan Volcanoes, there was another mantle plume in the same area, one that gave rise to the Comoros Islands between Africa and Madagascar, called the Mozambique channel. It was this plume that gave rise to a rift between Africa and Madagascar. The Comoros Islands remain volcanically active to this day. The Réunion plume is believed to be responsible for the separation of the Seychelles and India. Both the Comoros Plume and the Réunion plume are not as hot as they once were.

While the plume model to explain the volcanic activity has been popular for decades, it is not universally

the energies of Sun, causing a period of global cooling. The sulphur dioxide and carbon dioxide would later trap heat from the Sun, causing an extended period of global warming. Studies of sedimentary rocks in Meghalaya, show a troubled time for Earth, with widely fluctuating temperatures, and periodic acid rain falls. As a result of the volcanic activity, the Earth was inhospitable for almost 500,000 years. The formation of the Deccan Traps coincides neatly with the K-Pg extinction event, where 75 percent of all species alive on Earth died out. These included the dinosaurs. More information on the relationship between the formation of the Deccan Traps and the extinction event can be found in the January 2019 edition of Digit. 